

Day: _____

Date: _____

Course: Evolutionary
Algorithms

Instructor: Engr. Muhammad
Farooq

Submitted By:- Shayan Riaz

Reg No: F25ELE0319

Binary-coded Genetic Algorithm (BGA)

University of Engineering and
Technology Peshawar

Electrical Department
Artificial Intelligence

Task ONE Generation of Binary-coded Genetic Algorithm

Step No 01: Problem:

function $f(x) = x^2$
Where

Range: $-5 \leq x \leq 5$

Chromosome length = 5 per chromosome

Population size = 6

PC $\rightarrow$ Crossover probability $\rightarrow 0.8 = 80\%$

PM $\rightarrow$ Mutation probability $\rightarrow 0.1 = 10\%$

Step 02: Initial Population:

1 $\rightarrow$ 00101

2 $\rightarrow$ 01011

3 $\rightarrow$ 10100

4 $\rightarrow$ 11110

5 $\rightarrow$ 01101

6 $\rightarrow$ 10010

Step 3: Decoding Chromosomes:

Formula:-

$$\text{Lower} + \frac{(\text{Upper} - \text{Lower}) \times \text{DN}}{2^n - 1}$$

We have

$$-5 \leq n \leq 5$$

$$\text{Lower} = -5$$

$$\text{Upper} = 5$$

$\&$ N = number of chromosomes.

DV = discrete value for each chromosomes.

Formula

$$n = -5 + \frac{10}{31} \times DV$$

Chromosome	DV	X
00101	5	-3.39
01011	11	-1.45
10100	20	1.45
11110	30	4.68
01101	13	-0.81
10010	18	0.81

We convert binary $\rightarrow$ decimal
decimal $\rightarrow$ real value (X)

So we can use it in
the function.

Step 04:

Fitness Calculation:-

So

Fitness formula:

$$f(x) = x^2$$

After putting values in function:-

Chromosome	X	f(x)
00101	-3.39	11.49
01011	-1.45	2.10
10100	1.45	2.10
11110	4.68	21.90
01101	-0.81	0.65
10010	0.81	0.65

Smaller value = better Solution

So

Best = 01101 lowest value

0.65

Step 05:

Tournament Selection

Out-off given chromosomes
we have to make
pairs for mating pool

Pick a two solutions
randomly and compare
them.

Match	Compare	Winner
1	2 vs 4	2
2	1 vs 6	6
3	3 vs 5	5
4	2 vs 3	2
5	5 vs 6	5
6	1 vs 5	5

Mating Pool:-

2, 6, 5, 2, 5, 5

We select better solutions
by comparing two at a time.

Step 06 Crossover:-

Cut after 2 bits

Pair 1

01 011

10 010

→ 01010, 10011

Pair 2:-

01 101

10

01 011

→ 01011, 01101

Pair 3:-

01 101

01 101

→ 01101, 01101

Crossover mixes parents to create new Solutions.

Step 07 Mutation:-

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01010 → 01000

10011 → 10011

01011 → 01001

01101 → 01101

01101 → 01100

01101 → 01101

Mutation randomly changes bits to introduce diversity.

Step 08:

Decode offspring

Chromosome	Δv	x	$f(x)$
01000	8	-2.42	5.85
10011	19	1.13	1.27
01001	9	-2.10	4.41
01101	13	-0.81	0.65
01100	12	-1.13	1.27
01101	13	-0.81	0.65

We evaluate new solutions to see how good they are.

Step 09: Survivor Selection:-

Select best 6 ← 11001

01101 ← 11010

10010 ← 10110

01101 ← 10110

01101 ← 10110

01011

10011

We keep only the best solutions for next generation.

So

Crossover produced new solutions but not all of them were better than their parents.

Some offspring had higher objective values, showing that crossover does not always improve results.

Mutation introduced small random changes, which helped in exploring new

areas of the Search Space.
Survivor Selection is important
because it ensures that
only the best solutions
are carried forward to the
next generation, which
improves the performance
of the algorithm.
~~The explains behavior of~~
~~operations~~